

Data Modeling in DWH

Data Warehousing as a Practice



Intro

Data Modeling for DWHs

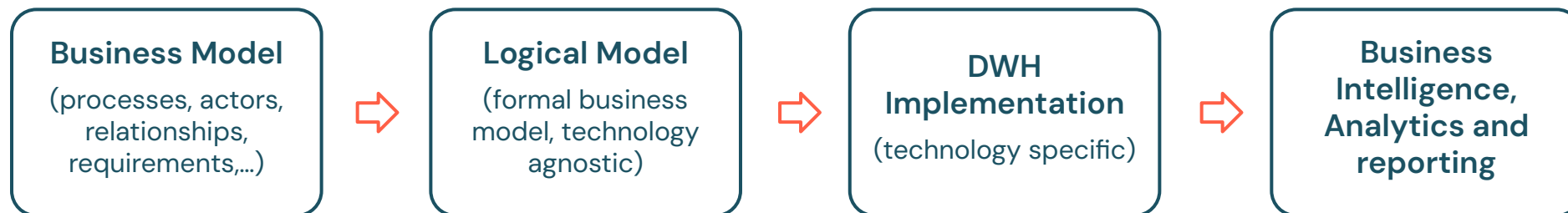
Why?

A DWH is used by business users to take business decisions.

As such, the data needs to be modeled in a way that it correctly represents the business (actors, processes, outcomes, ...) to ensure that insights and decisions based on the DWH are impactful.

Example approach

- Understand the business, its actors, relationships, processes, requirements
- Create a logical data model reflecting the organization's business processes and needs
- Ensure data in the data warehouse is of high quality, i.e. accurate, consistent, and well-organized
- Enable effective reporting and analysis to support for business intelligence, analytics, reporting, ...



Data Modeling for DWHs

How?

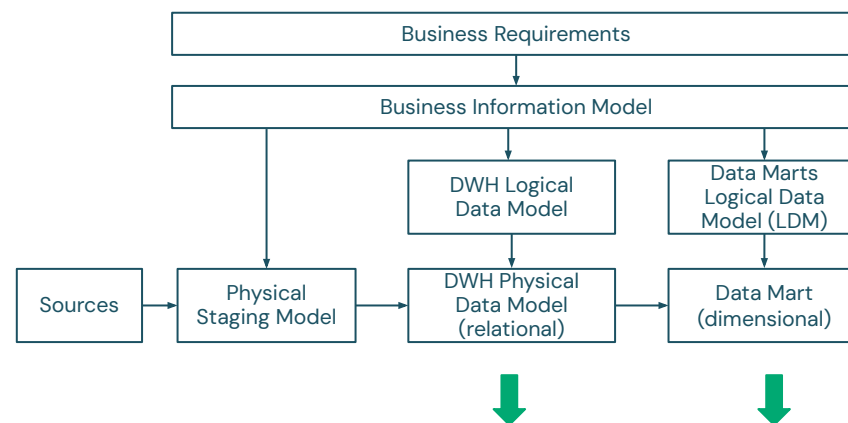
There are three main schools of thought for data warehouses:

- Top-down approach as defined by Bill Inmon (~1992)
- Bottom-up approach as defined by Ralph Kimball (~1996)
- Data Vault (v 2.0) as defined by Dan Linstedt (~2010)

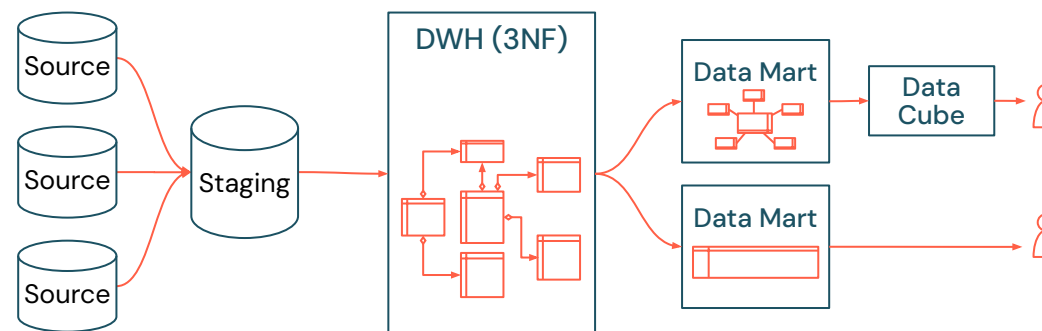
DWH modeling approaches – Inmon

- Relational Modeling with normalized data as the core of the DWH
- Data marts (often dimensional or denormalized models)

Process



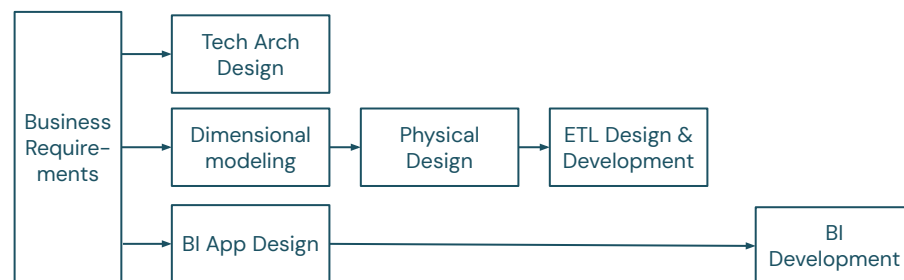
Logical View



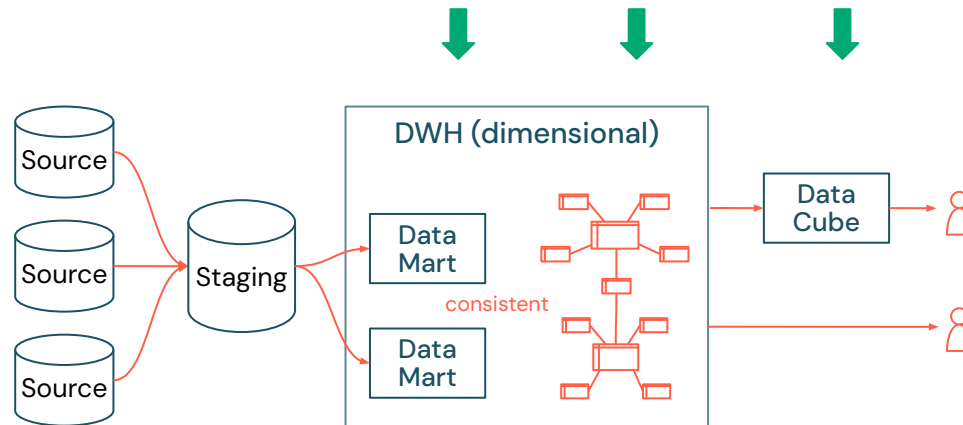
DWH modeling approaches – Kimball

- Dimensional Modeling
- Denormalized data model, built as a star or snowflake schema: central fact tables surrounded by dimension tables

Process



Logical View



DWH modeling approaches – Data Vault

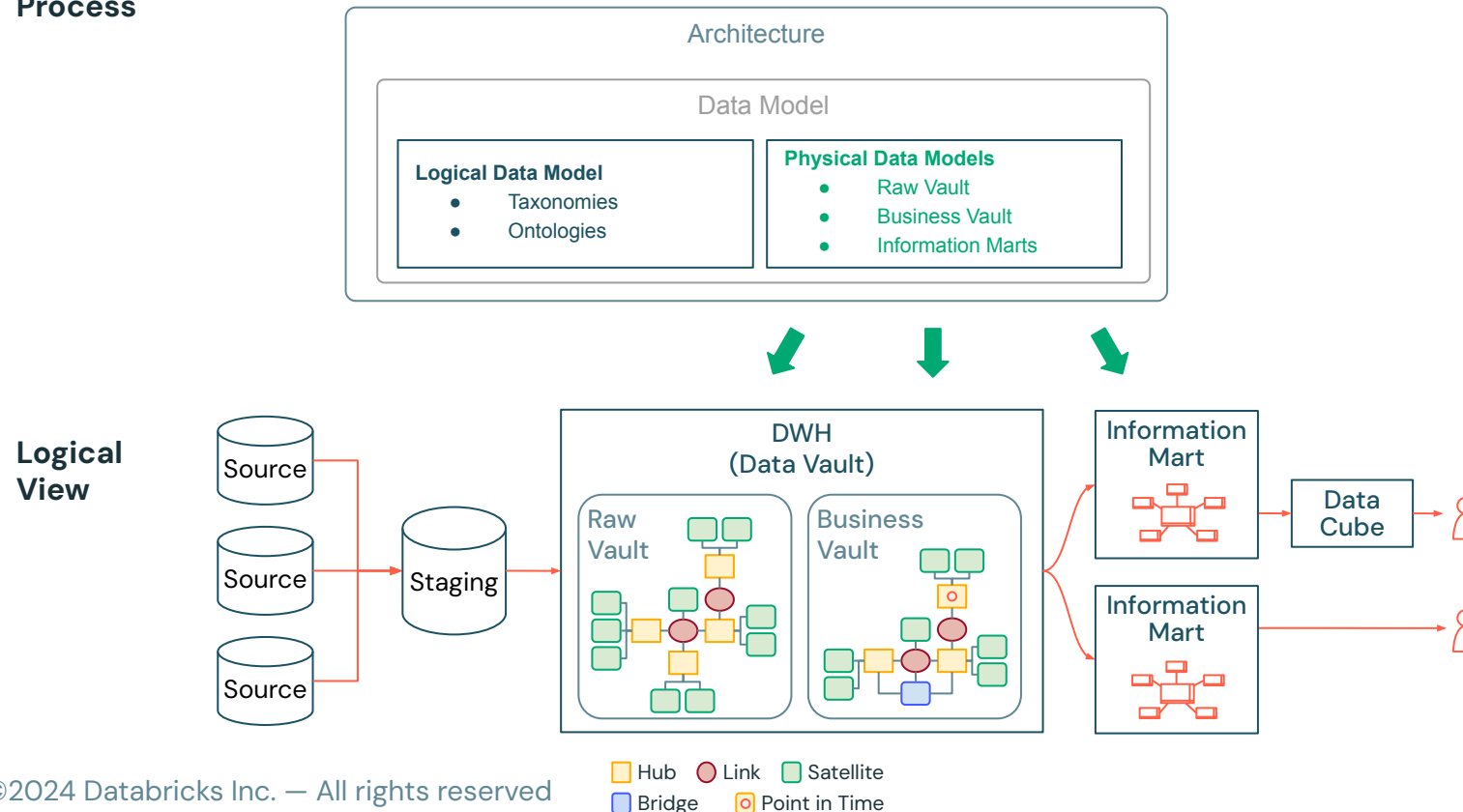
The Data Vault model is based on three basic entity types:

- **Hubs** separate core business concepts
- **Links** store relationships between business concepts
- **Satellites** store the attributes of a business concepts or relationships

The Data Vault model is split into

- **Raw Vault / Raw Data Vault:**
 - stores unaltered, granular source data.
 - immutable, historical record of all the data in the Data Vault.
- **Business Vault / Business Data Vault:**
 - sparsely modeled DWH based on Data Vault design principles
 - data is modified according to business rules or requirements
- **Information Marts**
 - Like Data Marts
 - Dimensional model based

Process



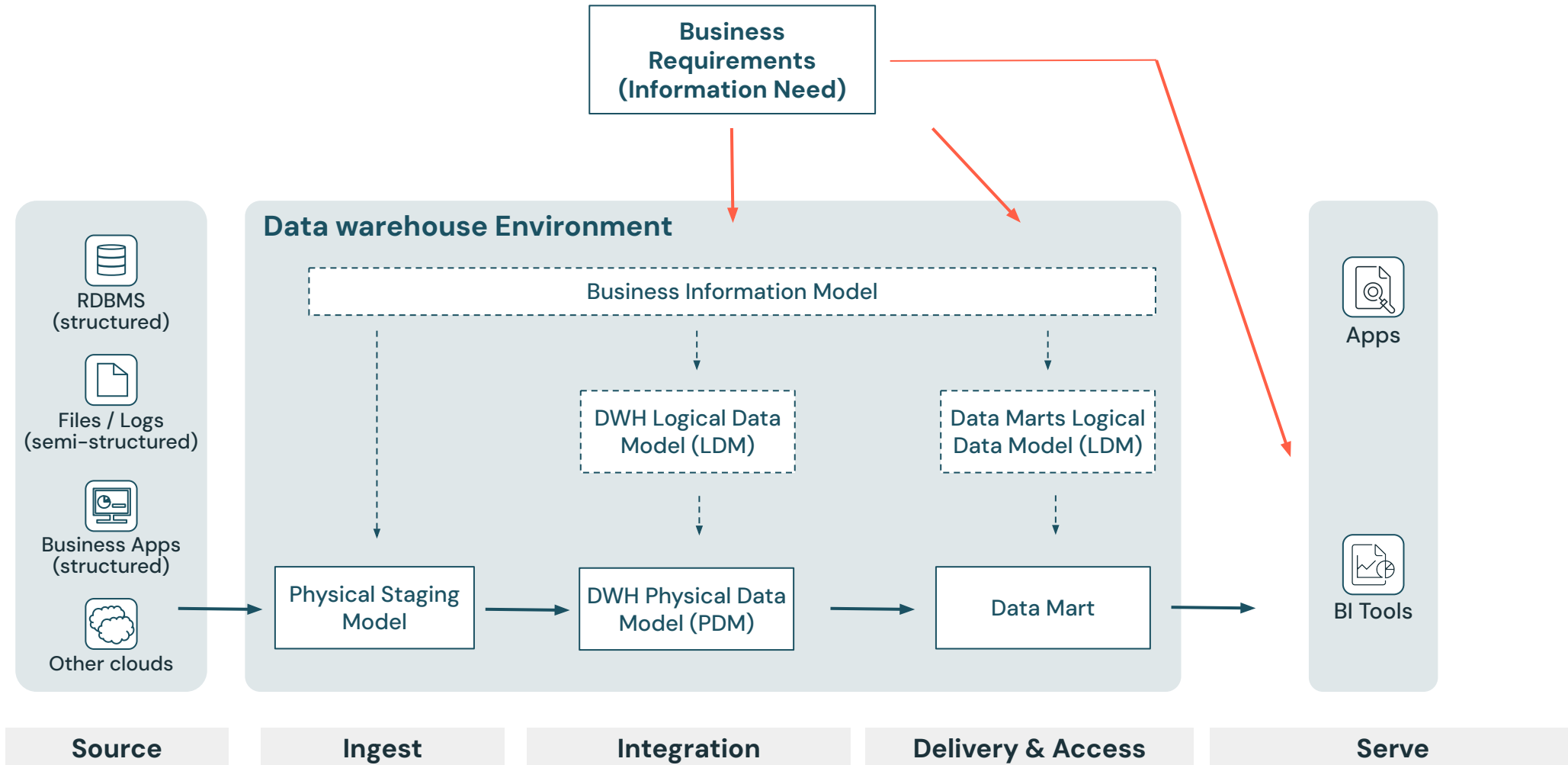
Conceptual data modeling

Context

We describe the conceptual approach to data modeling based on the terminology of the Inmon approach.

However, conceptually, the logical model for example can also be an ontology and taxonomies as used in Kimball and Data Vault

Data warehousing – Purpose of modeling



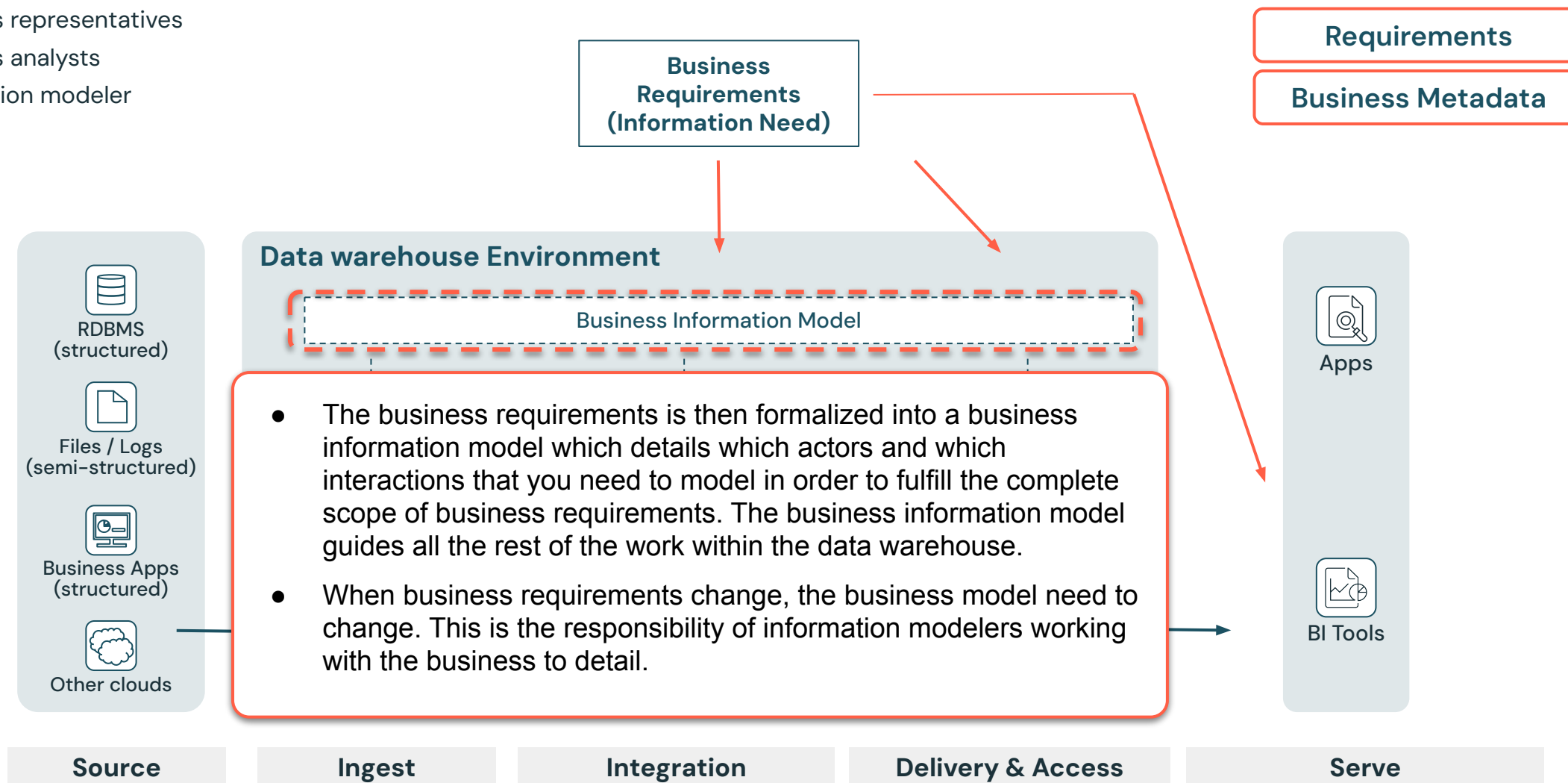
Business Requirements

-  Business representatives
-  Business analysts



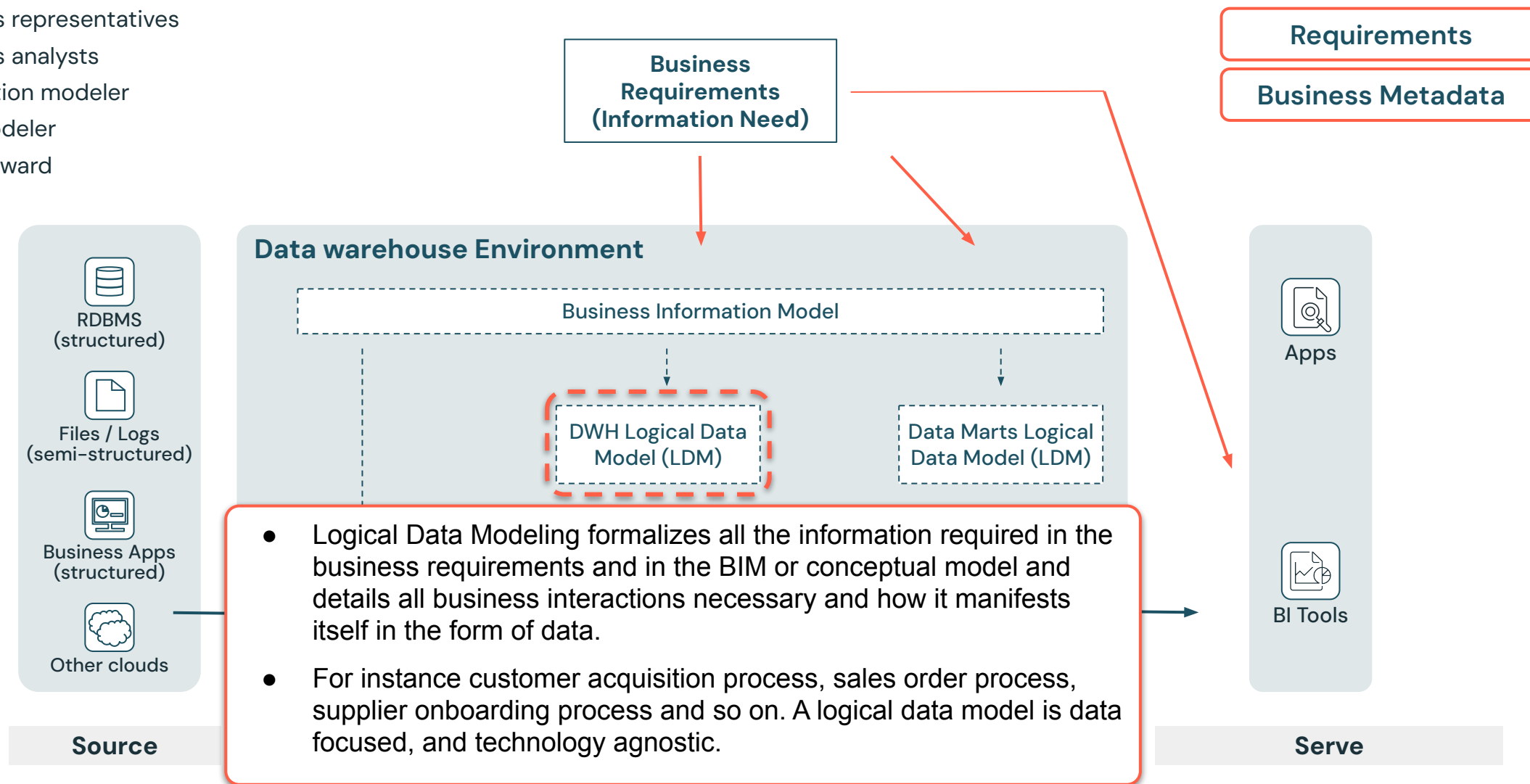
Business Information Model (BIM)

-  Business representatives
-  Business analysts
-  Information modeler



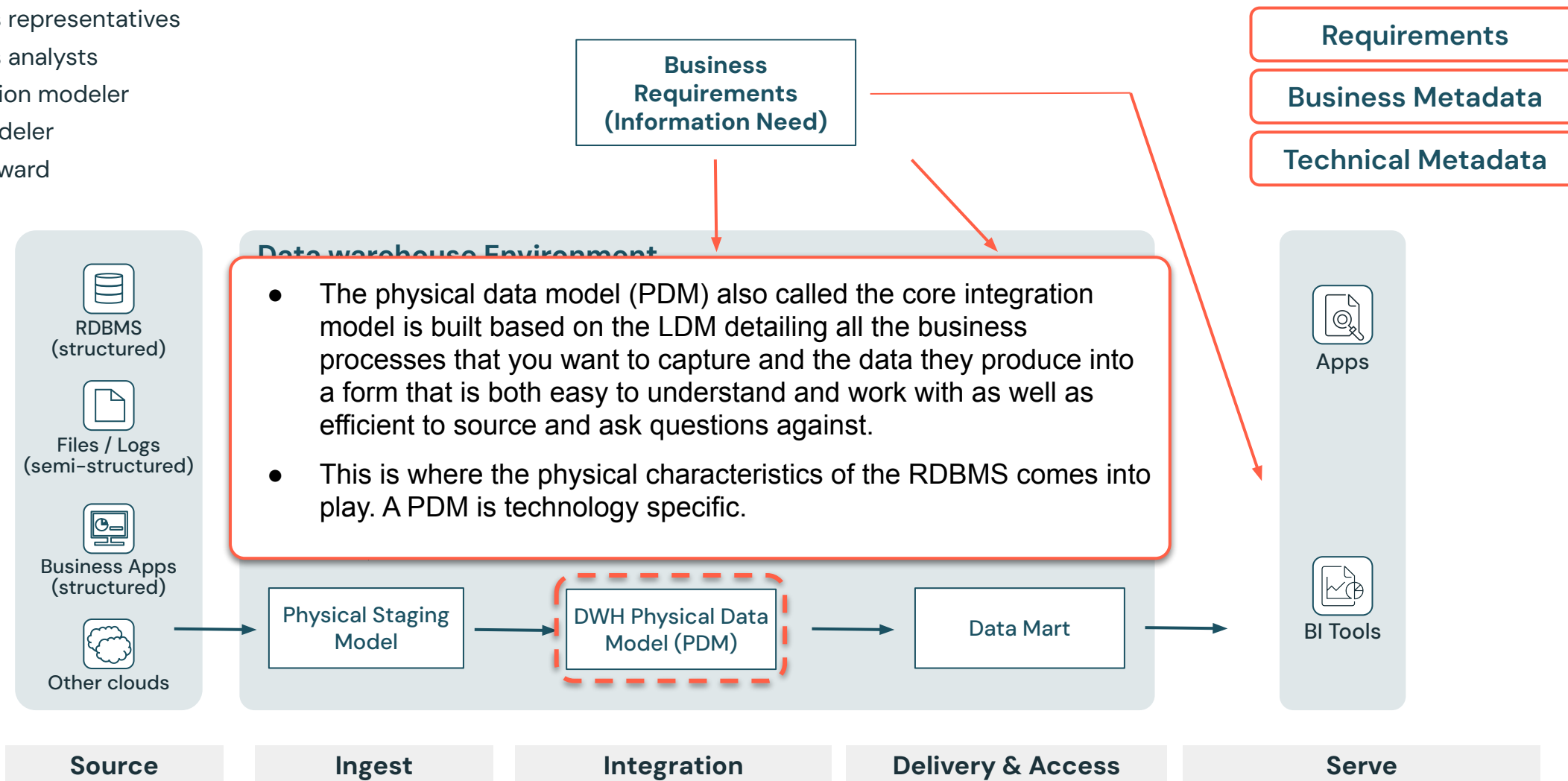
Logical Data Model (LDM)

-  Business representatives
-  Business analysts
-  Information modeler
-  Data modeler
-  Data steward



PDM (Core integration Layer)

-  Business representatives
-  Business analysts
-  Information modeler
-  Data modeler
-  Data steward



Data Marts

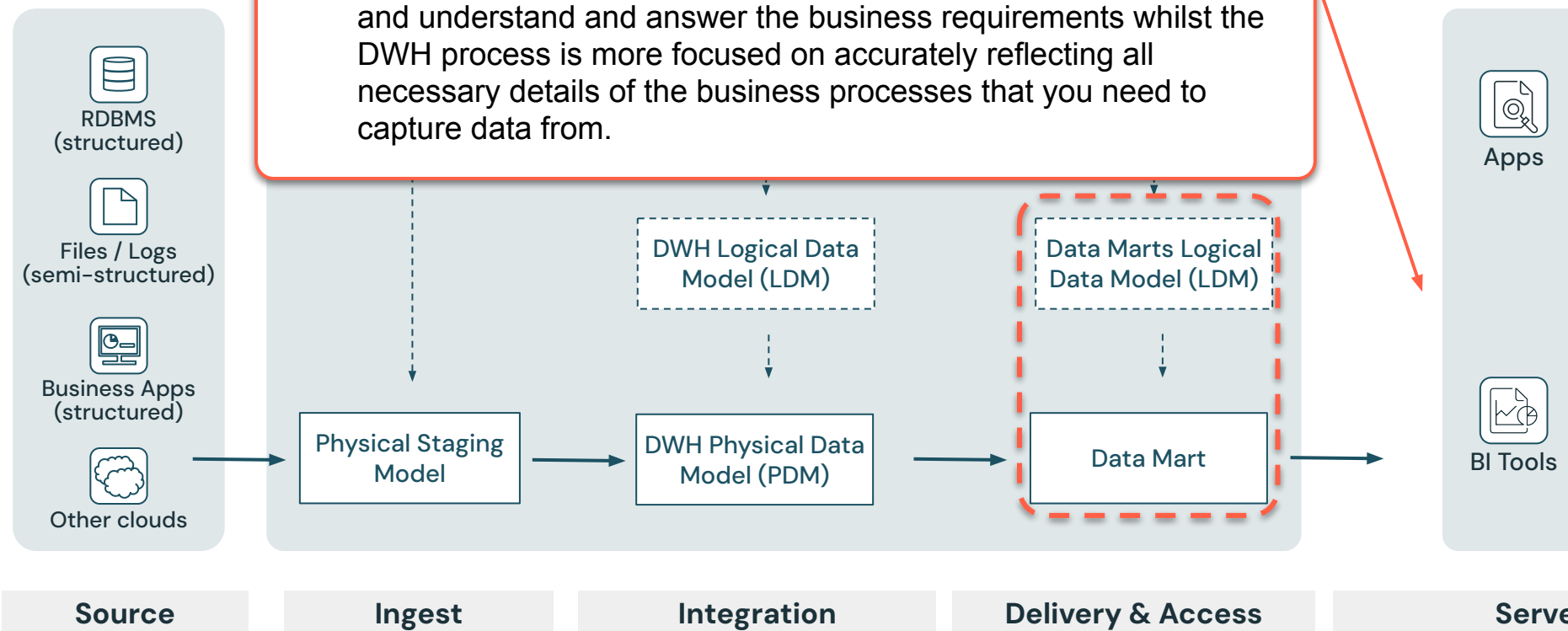
-  Business representatives
-  Business analysts
-  Information modeler
-  Data modeler
-  Data steward
-  Data analyst

- In essence the data mart process is very similar to the DWH core process but works on business process meaningful groupings and entities.
- You can basically see it as simplifications to make it easy to access and understand and answer the business requirements whilst the DWH process is more focused on accurately reflecting all necessary details of the business processes that you need to capture data from.

Requirements

Business Metadata

Technical Metadata



Principles

1. Be true to the business processes you model.
2. Apply business rules as late as possible in the chain.
3. Avoid cross dependencies up to the point you absolutely need them.
4. Break monoliths in smaller components to make them easier to source, manage, scale

Key takeaways

- Much of the data warehousing process is technology independent and will still apply in the context of a lakehouse.
- The DWH practice requires specific tools outside of the Lakehouse
- The core integration layer (PDM) is primarily used to ensure consistency, correctness of representing the data (single source of truth) and as such start and end points for ETL processes.

Modeling a Data Warehouse – Inmon

Building a DWH – the process

Models are front and center of building a data warehouse

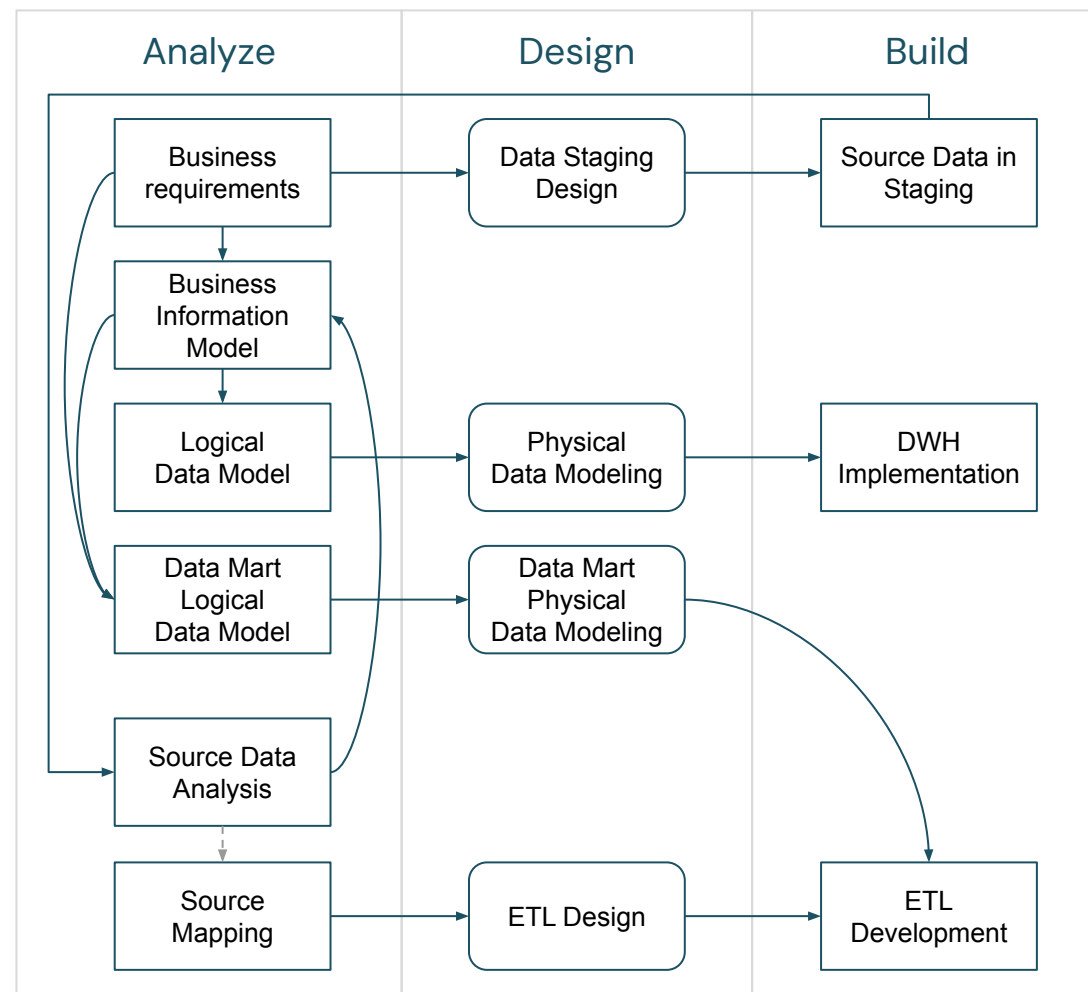
- **Business Information Model (BIM)**
Modeling actors, their relations and how they make business together (“how the business works”)
- **Logical Data Model (LDM)**
Model of the data that is associated with the BIM
- **Physical Data Model (PDM)**
Implemented data model derived from the LDM

Models describe the business world and its relationships, i.e. depicts the business processes within the organization. They generate the business context required to create business information from the data and store it accordingly.

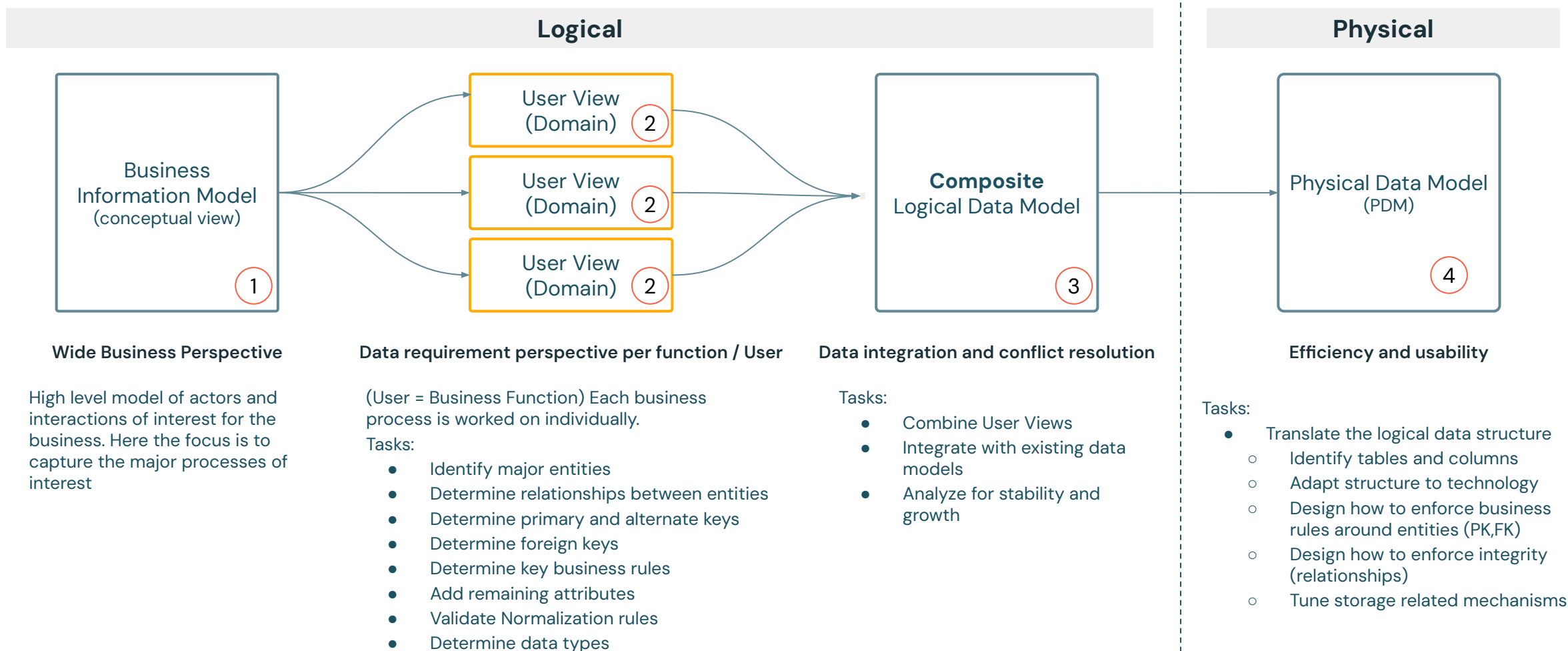
For this process

- **Analyze** is technology agnostic
- **Design** is impacted by technology (e.g. impacted by non functionals like performance, cost/benefit, ...) and understandability for consumers
- **Build** is using the actual technology to implement the physical model and the ETL processes

Simplified DWH Process



Data Modeling work-process (Inmon)



⇒ This is an iterative ongoing process across the warehouse lifecycle where information captured in later steps may inform prior steps.

Entities, and attributes in LDM

- **Entity** = Person, place, thing or concept about you wish to record facts
- **Attribute** = a non decomposable atomic piece of information describing an entity
- **Non Decomposable** = Smallest unit of information you will want to reference
- **Business rules** = specifications that preserves the integrity of the LDM by governing which values attributes may assume
- **Business rules** (two categories):
 - Key business rules – The identification of unique records
 - Domain business rules – Validation of attribute values

It is useless to understand the structure of information within the business without understanding the rules pertaining to the integrity of the information.

Optimal LDM

1. **Structural validity** – Consistency with how the business defines and organize information
2. **Simplicity** – Ease of understanding
3. **No redundancy** – No extraneous information
4. **Shareability** – Not specific to one solution, usable by many
5. **Extensibility** – Ability to evolve with minimal effect on existing base
6. **Integrity** – Consistency with the way the business uses and manages information values

Handbook of relational database design

(Candace C. Fleming, Barbara von Halle)

Dated, old but the sections about logical data modeling is still one of the best

Normalization and the Databricks Platform

- Databricks' parallel engine (Apache Spark) is tremendously good at scanning and processing large volumes of data since these steps can be done in parallel over N number of workers.
- Joins for the most part lead to exchange of data between workers through serialization and deserialization. But by utilizing modern features such as **liquid clustering**, **predictive optimization**, **deletion vectors**, and **gathering of statistics** one can dramatically reduce the impact of normalization.

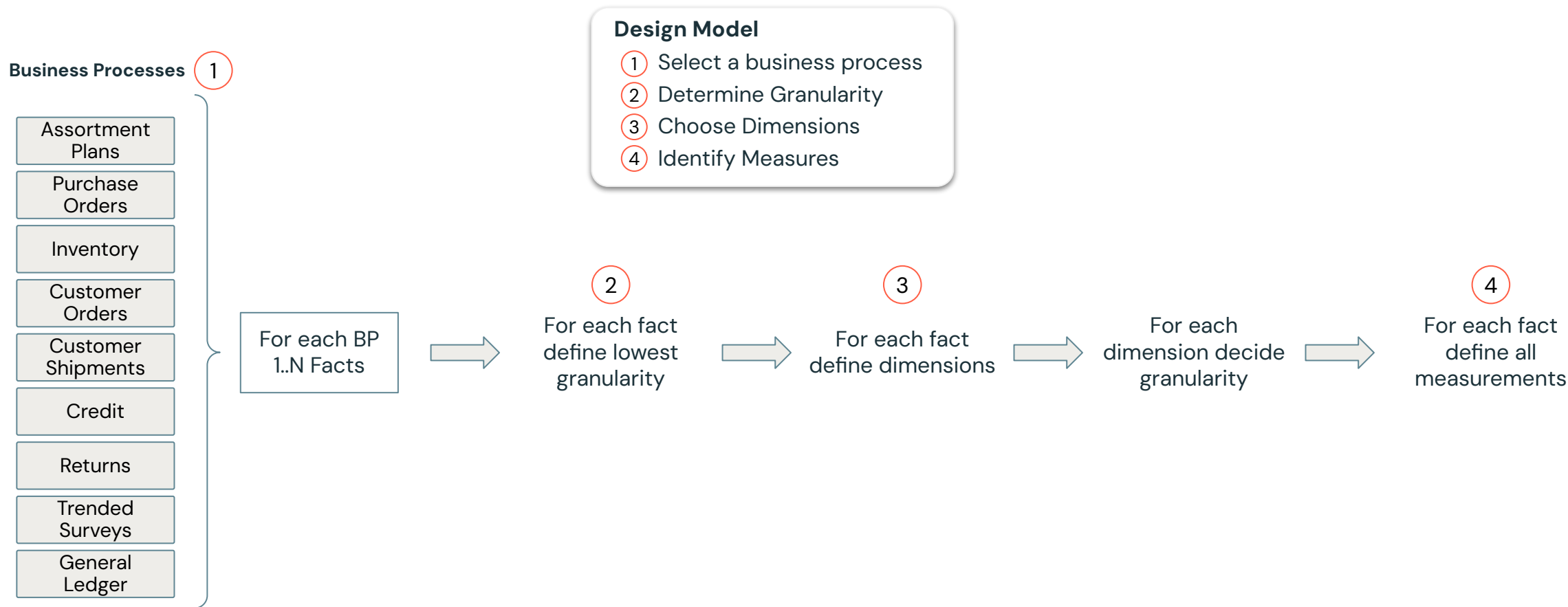
Modeling a Data Warehouse – Kimball

Dimensional modeling according to Kimball

Fundamental Concepts

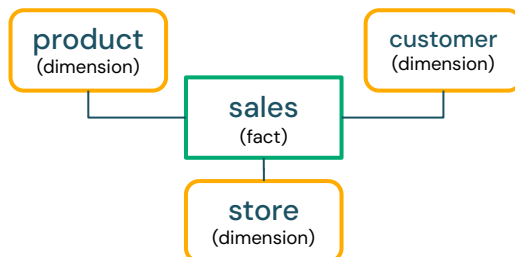
- 1. Gather Business Requirements and Data Realities**
Understand the needs of the business, decision making processes, ...
- 2. Collaborative Dimensional Modeling Workshops**
Design in collaboration with subject matter experts and data governance representatives from the business
- 3. Four-Step Dimensional Design Process**
Work on the four key decisions, see next slide
- 4. Business Processes**
Business process events generate or capture performance metrics that translate into facts in a fact table.
- 5. Grain**
Deciding on the grain of fact tables (i.e. each row) is a prerequisite to consistency
- 6. Dimensions for Descriptive Context**
The “who, what, where, when, why, and how” context surrounding a business process event
- 7. Facts for Measurements**
Fact tables correspond to a physical observable events. Rows in fact tables have a 1-to-1 relationship to a measurement
- 8. Star Schemas and OLAP cubes**
Typically apply multi-dimensional modeling (star schemas) and, if needed, create cubes for OLAP tools
- 9. Grace Extensions to Dimensional Modeling**
Follow techniques that enhance the flexibility and adaptability of dimensional models when data relationships change

Example Logical Design of a Retail customer



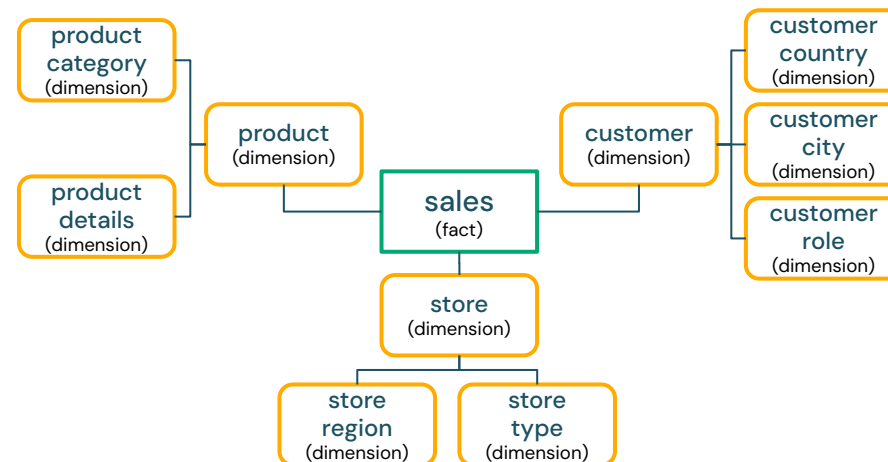
Multi-dimensional data models

Star schema



- Fact table contains business "facts" (like transaction amounts and quantities)
- Dimension tables contain information about descriptive attributes and are typically denormalized
- Star schemas enable users to slice and dice the data, typically by joining two or more fact tables and dimension tables together

Snowflake schema



- Fact table as with star schema
- Dimension tables are broken down into subdimensions
- Dimensions are normalized
- Simple data model enforcing data quality, with fast retrieval
- Higher setup and maintenance efforts

Denormalization and the Databricks Platform

Denormalization

- The advent of columnar storages such as Delta Lake reduced the need of strict normalization since having multiple columns in the same table does not induce a cost of having to scan the complete row any more.
- Denormalization almost always means duplication of data at some level, but due to storage compression mechanisms and the ability to do good filtering, the impact of denormalization is limited.
- Together with the ability to store row formats in columns means that you can have tables pre-joined but isolated in separate structs and thereby can be treated as individual tables, or as a pre-joined result.

Modeling a Data Warehouse – Data Vault

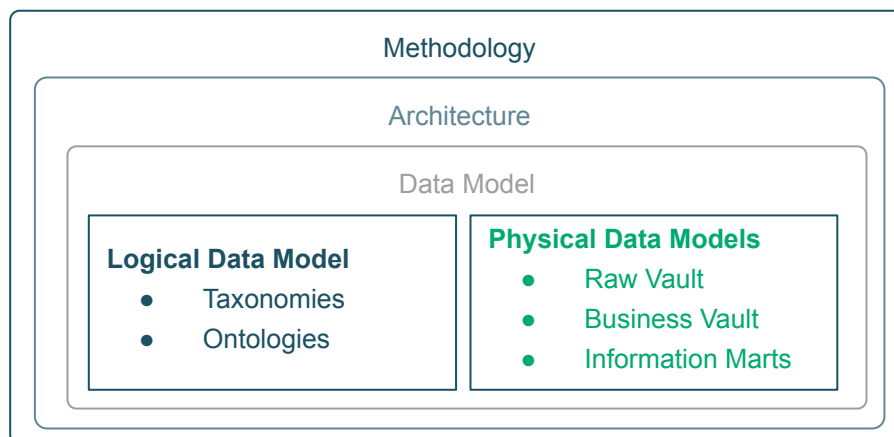
Data Vault 2.0

“Data Vault Modeling was, is, and always will be about the business”

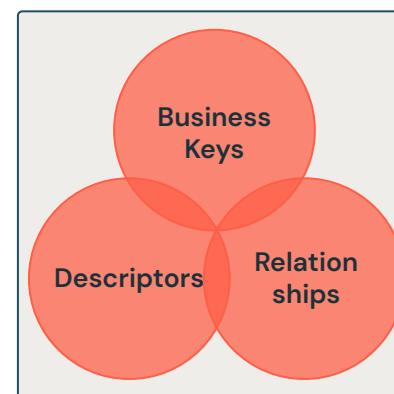
Dan Linstedt

Theoretical Structure

Data Vault 2.0 is a methodology covering Data Architecture and Data Modelling



Logical Data Modeling in Data Vault captures



through taxonomies and ontologies

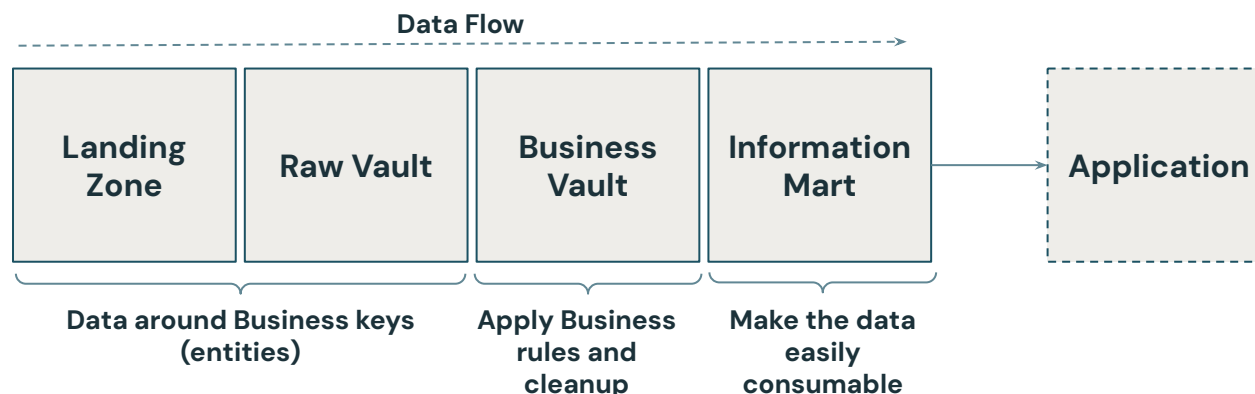
Business keys – Major Entities

Relationships – Business rules

Taxonomy – A formal hierarchical structure of business elements representing a business domain

Ontology – A business representation of the business object state and relationships in the form of triplets (subject-predicate-object) and axioms

Physical Structure



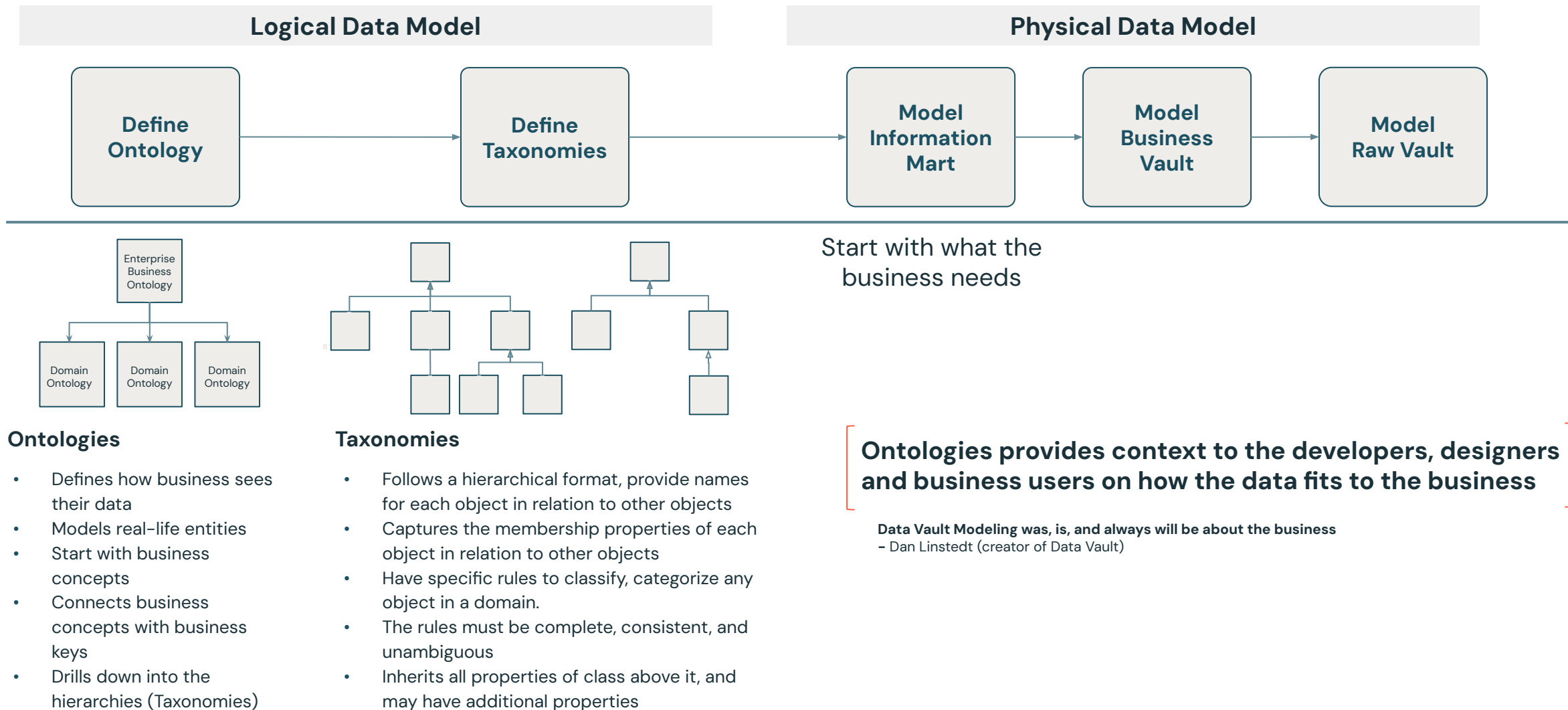
Data Vault is a response to the challenges identified to use 3NF modelling (Inmon) and Dimensional modelling (Kimball).

The primary objection of using 3NF is that it is on a too granular level and business rules have a ripple effect across the model, leading to a cumbersome model to maintain and handle.

The primary objection against Dimensional modelling is that it leads to dimension explosion, very hard model to evolve and maintain.

Data Vault claims to have solved many of these problems by lifting out business rules to a separate layer.

Data Vault work-process



Benefits and impacts of the three approaches

Evaluating the three modeling paradigms

Examples – non exhaustive

- **Ability to change**
 - Big impact on Inmon models when business process changes => higher effort and duration
 - Changed business (especially significant changes) can break the basis of a Kimball model => higher effort and duration
 - Data Vault structure should make it easier to react on business changes => lower effort
- **Complexity**
 - Inmon leads to very complex ETL and load dependencies that need to be handled through load flow optimizations, or extra ETL jobs to ensure the model is consistent.
 - Kimball dimensional models can be very hard to populate since you have to ensure consistency with the dimensions. Dimension logic can be hard, particularly slowly changing dimensions type 2,4, and 6 can be very tricky.
 - Data Vault has between 3.5x and 5.5x more objects than a pure 3NF DW which has an impact on ETL. But the ETL is simplified and can for the most part be run in parallel, and can easily be automated.
- **Robustness**
 - Inmon models can easily break due to changes in business processes and business rules. That can have a massive impact on the model.
 - Kimball is the simplest model to understand, but many changes basically means remodelling parts.
 - Data Vault: Most changes can be compartmentalized to a specific layer
- **Methodology**
 - Waterfall (Inmon, Kimball) vs. agile (Data Vault, Kimball)

Advantages of the three approaches

“Unbiased” attempt based on LLM summarization

Kimball (dimensional)

Optimized for Reporting and Analytics:

Dimensional models are designed specifically for efficient querying and reporting. They provide a clear structure for business users to understand.

Easy to Understand:

Business users find dimensional models intuitive due to their star schema or snowflake schema representation.

Well-Suited for Smaller Projects:

Quicker to implement for smaller-scale data marts.

Inmon

Normalized Structure:

Inmon models follow a normalized approach, reducing data redundancy.

Single Source of Truth:

The data warehouse is designed as a single integrated repository.

Well-Suited for Large Enterprise Data Warehousing:

Recommended for large-scale data integration.

Data Vault

Operational Flexibility:

Data Vault allows you to stay close to the source data, making it auditable and scalable.

Easier to Add New Sources:

Data Vault is flexible and accommodates new data sources seamlessly.

Historical Data Tracking:

Inherent support for historical data.

Source: bing copilot 2024-02-07

“Compare “Dimensional Modeling” (Kimball), Inmon modeling, Data Vault v2.
What are the advantages and what are the challenges”

Challenges of the three approaches

“Unbiased” attempt based on LLM summarization

Kimball (dimensional)

Data Redundancy:

Dimensional models may have some data redundancy due to denormalization, which can lead to maintenance challenges.

No Single Source of Truth:

Data marts are organized around business areas, which can result in multiple versions of the same data.

Inmon

Complexity:

Inmon models can be complex to implement and maintain.

Slower Query Performance:

Normalized structures may require more joins, impacting query performance.

Less Intuitive for Business Users:

Business users may find the normalized structure less intuitive.

Data Vault

Not Ideal for Analysis & Reporting:

Data Vault may not be the best choice for direct analysis and reporting; you might still need dimensional modeling for virtual data marts.

Complexity:

Data Vault models can become complex, especially when directly populating data marts from them.

Joins and Recursions:

Populating data marts from Data Vault models can lead to complex joins and recursions.

Source: bing copilot 2024-02-07

“Compare “Dimensional Modeling” (Kimball), Inmon modeling, Data Vault v2.
What are the advantages and what are the challenges”

Common challenges with data modeling

A reason why “no modeling” seems to be attractive

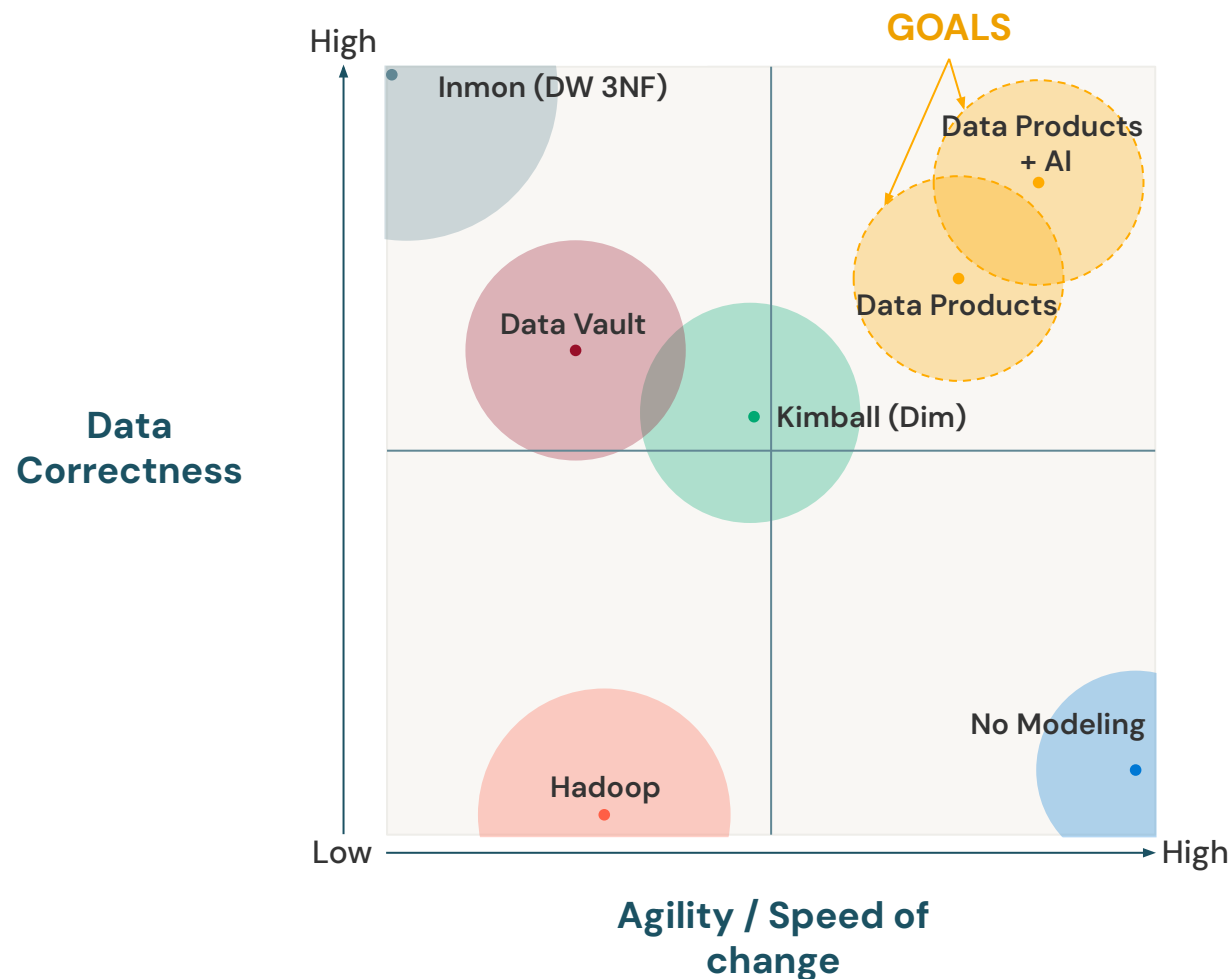
Many organizations find it challenging to handle the life cycle around data models. The challenges comes in the form of people, process and technology.

- To maintain a database you need DBAs or data engineers
- To model databases you need data modelers
- To do a “correct” data model you need access to the business

For most organizations the overhead of having data modelers talking to the business and the time it takes to introduce a change is simply not worth it since it affects the ability to adapt to new conditions.

But cutting corners in this process has side effects: Data correctness, and potentially data quality goes down. We are trading correctness for speed and agility in the data process.

Correctness vs ability to support change



There is a constant tradeoff between agility and correctness, but with the lakehouse you can selectively decide what matter most for the specific use case in question.

Note: This represents our point of view and is most probably very controversial

Data Categories

Categories of data

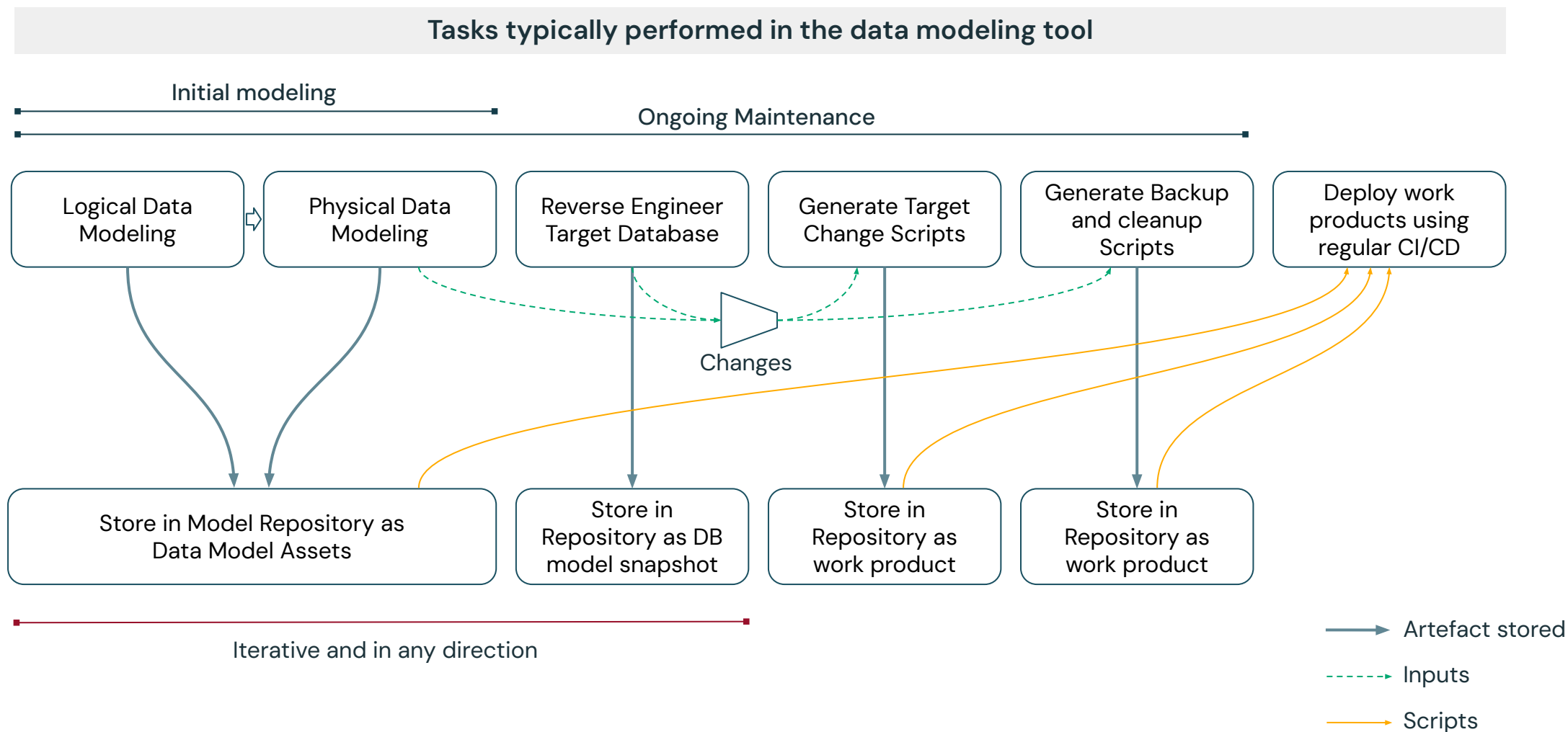
Type of Data	Characteristics	Challenges
Master Data	Data which is agreed upon and shared across the enterprise <i>Examples: Customer, Product, etc.</i>	• Agreements on master data entities can be hard to achieve, especially in large organisations • Often disputes/disagreement on definitions between different units/departments
Reference Data	Subset of Master Data that refers to data that defines the set of permissible values or standardized definitions to categorize or describe other data <i>Examples: Country Codes, Currencies, etc.</i>	• The huge amount of reference data can make it hard for organisations to handle it • Different levels (local code lists and common code lists) need to be kept aligned • Localisations especially in case of multinational companies
Transactional Data	Business–event oriented data captured from transactions <i>Examples: Order (which customer bought which product at which time), Withdrawals (which amount was withdrawn by whom at which time at which location), etc.</i>	

The special role of MDM and RDM

- Every DWH contains master data, reference data and transactional data.
- The DWH is not the true master of curated master data and reference data. There typically is a separate practice called Master Data Management (MDM) and Reference Data Management (RDM, a subset of MDM) involved. However, for organizations lacking MDM and RDM practices, the DWH tends to be seen as the true source of master data and reference data
- RDM and MDM are not handled in this deck

Data Modeling Tasks

Data Modeling tasks during Implementation



Tools

Modeling & Automation tools

Practices

Modeling & model deployment

[SQLDBM](#)

[Erwin Data Modeler](#)

Model driven Automation

[VaultSpeed](#)

[WADE Insight](#)

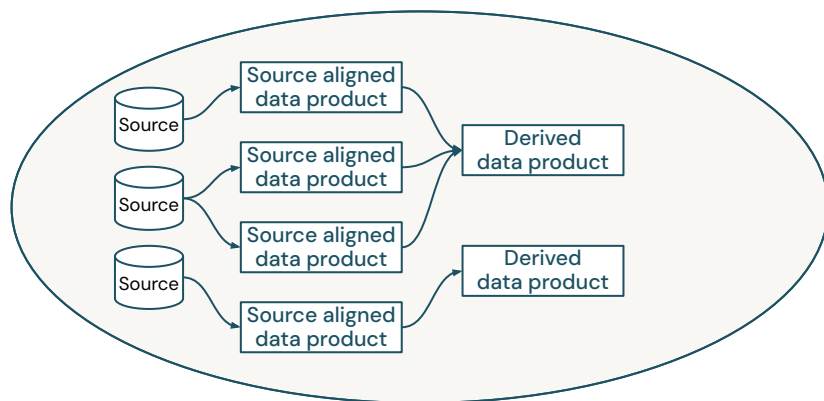
[Agile Data Engine](#)

...

Data products and the DWH

Data products to combine different worlds

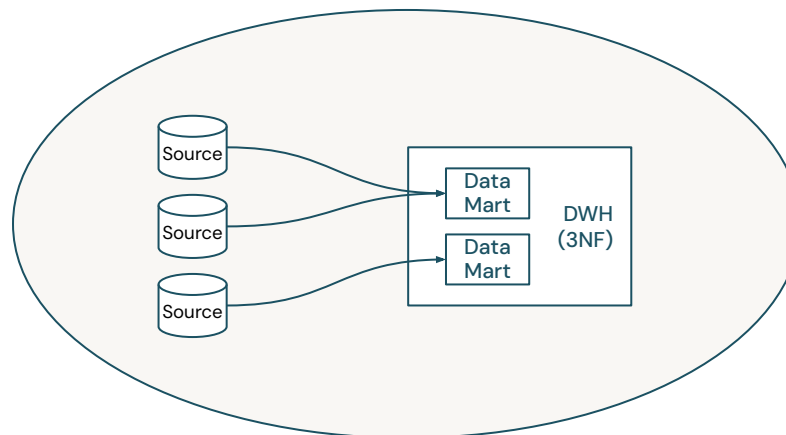
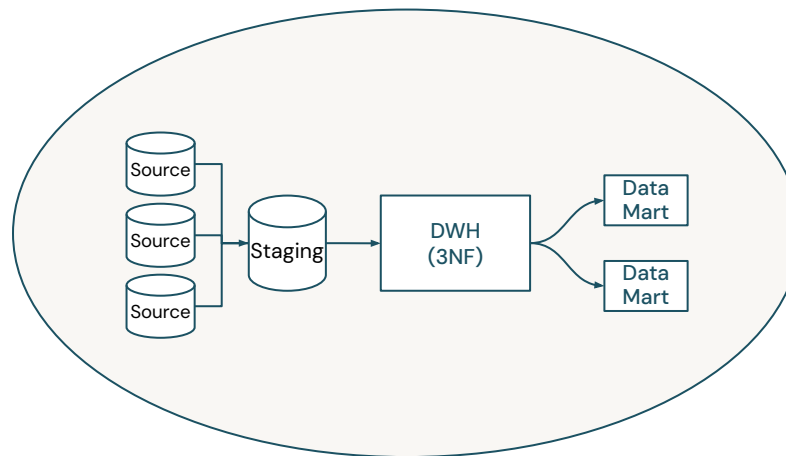
Domain having standardised on the data products paradigm



Data product (facade)

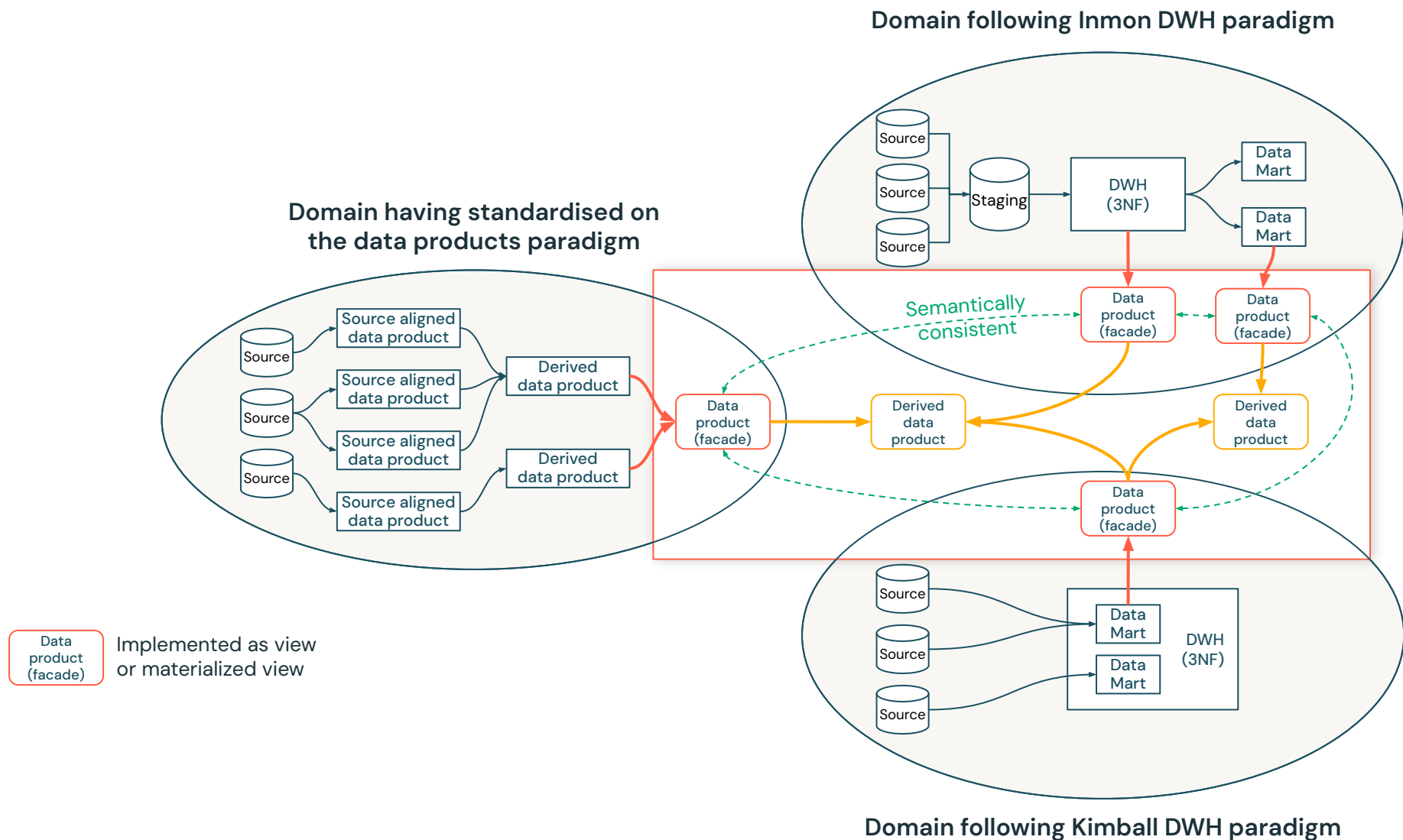
Implemented as view or materialized view

Domain following Inmon DWH paradigm



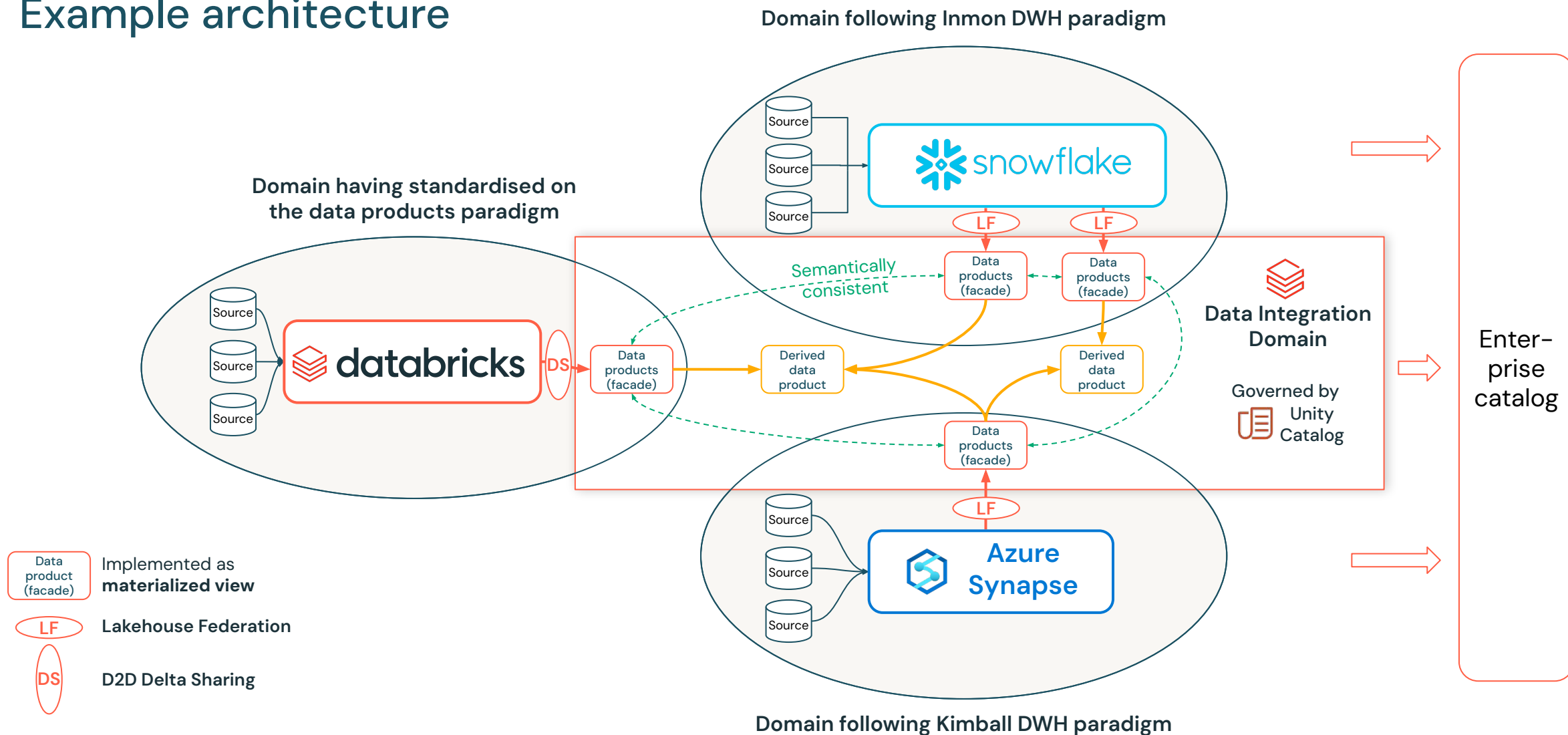
Domain following Kimball DWH paradigm

Data products to combine different worlds



Integrate into UC (LH Federation, D2D Sharing)

Example architecture

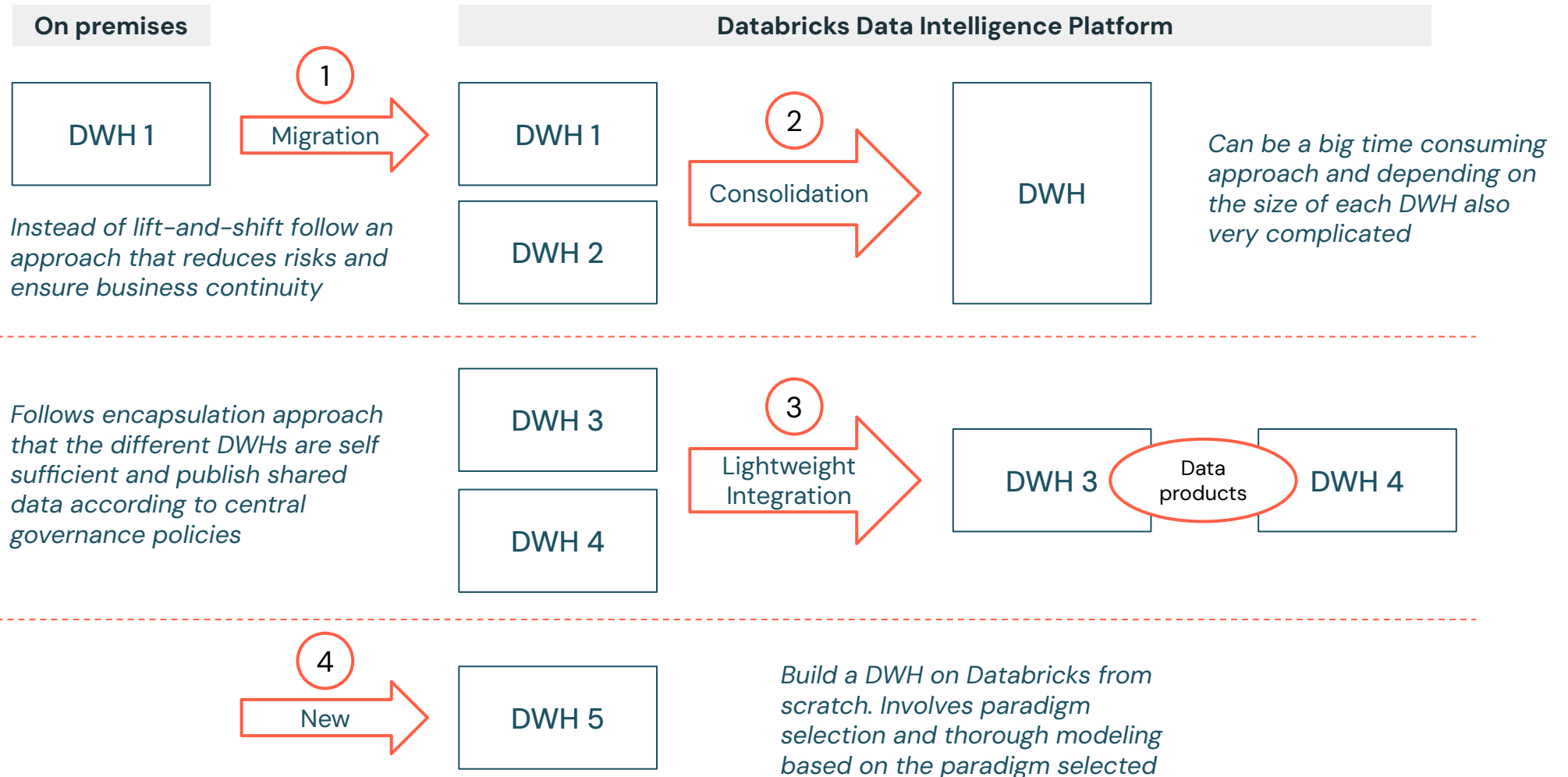


Example architecture



Data Warehouse Migrations

DWH project types in cloud



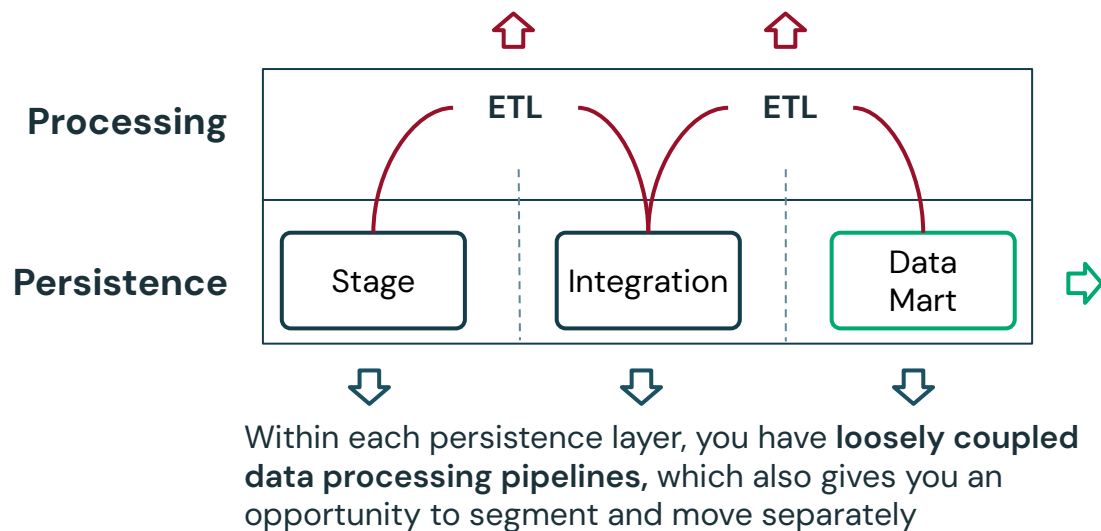
Stepwise migration of data warehouses

Evolve rather than make a big bang

- By **incrementally migrating** a data warehouse, you can move and evaluate each step to see if you have met the objective of each step in the transition.
- If the plan is **sufficiently segmented**, and **stable interfaces have been defined** with clear goals of each step, the transition will be easily managed.

ETL is typically expressed in external ETL tools. You have many options depending on the goal:

- Run as is against a new target (ETL tasks run in the ETL tool)
- Pushdown into database (ETL tasks run in the target data warehouse)



The Data Mart supports usage of the data

- Data Marts can be moved first to prove the value of the new data platform to business.
- This typically offloads the existing platform to have more processing capacity available for supporting other tasks.
- Business reassurance that the move is valuable.

Develop towards stable interface definitions

Software engineering practices such as well defined interfaces, encapsulation, and reuse also apply to building data projects

Data warehousing has a tendency to be hard to build in an agile fashion since everything starts with data. By **defining clear interfaces**,

- you can run parallel teams building towards different interfaces.
- change management is handled through versioning of the interface.
(this means that the impact of a new version needs to be assessed, and handled whenever the interface changes. Typically this is a much quicker process than waiting until the data layer is complete, before building the next step in the process is possible)
- you gain ability to build test cases, and coverage metrics.

Generalization and flexibility to the right degree

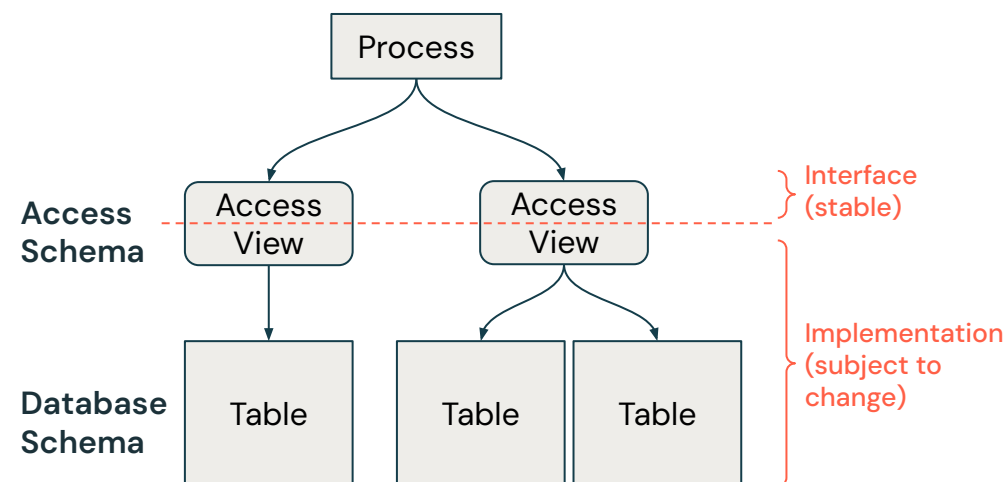
- Many times completely generalized metadata driven solutions reach a breaking point.
- Building such frameworks is often driven by the ambition to make things simpler, but the end result is many times a black box solution that is hard to maintain and understand.
- Transparency, understandability, maintainability, testability and predictability is core to any well defined data process. Code generators are great, but it is even better to generate code that you can test, validate and treat as an asset in a well defined CI/CD process rather than having a code generator in production.

Stable interfaces

Example of an implementation via view layers

View layers

- The easiest way to define stable interfaces in databases, is to **ensure all access goes through a view layer** with all the columns explicitly listed in the right order.
- View layers are there in case you need to change anything behind the view layer: **You can then still guarantee the interface** (provided that is what you want of course).
Let's say for instance you provide a table for customer data to consumers, but then decides to reorganize the data layer in relation to customer data. By using a the view layer I can ensure the same interface for consumers is provided by altering the internals of the view to capture data in accordance with the new structure but keep the output format the same.
- This uses a core concept that already exists in most data warehouses: **Never access tables directly**. If you allow for accessing tables directly you have a strong coupling between the consumer and producer. By have a layer of redirection in between you can always handle change of the data layer in many different ways.



```
CREATE VIEW experienced_employee (  
    id int ,  
    name string  
)  
AS SELECT id, name  
FROM all_employee  
WHERE working_years > 5;
```

The SQL code demonstrates the implementation of a stable interface. The **CREATE VIEW** statement defines the **Interface (stable)** with columns `id` and `name`. The **AS SELECT** statement defines the **Implementation (subject to change)** by selecting data from `all_employee` where `working_years > 5`.

DWH Migrations – Data Marts

Optimizations during migration need to support the business purpose

- Data Marts are the backbone of serving the business in a DWH, and as such efficiency is fundamental.
- When migrating existing Data Marts, efficiency in the new platform needs to be a consideration. However, for familiarity and experience reasons of the users, one might decide to not change it and accept the technical inefficiency.
- However efficiency and technological considerations are more important for layers immediately serving the business.

Further reading

Further readings

- [Comparative study of data warehouses modeling approaches: Inmon, Kimball and Data Vault](#)

Good assets for modeling

There is a very good IDERA ER/Studio tutorial on youtube detailing how to do logical data modeling, and the whole process around data modeling.

Even if the specific tool may not be interesting, it is very good as basic understanding how modeling is done and generally the type of tools the modeling professionals are using.

Appendix

DWH Modeling assets

Work tracks

Logical (Technology independent)

- Business Requirements
- Business Information Model
 - Conceptual Model (core entities)
- Logical Data Model
- Data Marts Logical Data Models

Manifested in:

- Business Models
- Business Glossaries

Physical (technology dependent)

- PDM (Core Integration Layer)
- Physical Data Mart Models

Manifested in:

- DDL scripts

DWH Asset Management

Logical (Technology independent)

Tools

- Specific DWH tooling for modeling and asset management of models are often required.

Example products:

- Erwin, IDERA ER/Studio, and others

Physical (technology dependent)

Tools

- Physical models are often generated from DWH tooling in the form of DDL scripts or applied directly from a tool towards a RDBMS engine.
- DDL scripts can be checked in and maintained in solutions such as GitHub but often the link to the logical model is broken if maintained outside of the DWH tooling for modeling.

How to build an LDM

Questions:

- 1) What are the major information objects or concepts of interest to be modelled
Example: Customers, service requests, service calls, bills, payments,...
- 2) What detail information items characterize those information objects
- 3) How does the information objects, and concepts relate?
(*Hint: Thorough understanding of the business processes are required*)

Example:

- *Customers make service requests, and bills are sent to customers requesting payments for service calls.*

Relational Data Modeling

Relational Data Modeling captures the business context from business processes. This is distilled to entities, and relationships between entities and business rules for attributes.

Relationships expressed are one of three types

- Existence relationships (parent has children)
- Functional relationships (professor teaches students)
- Event relationship (customer places order)

And the cardinality and ratio is also derived from what is “plausible” in the business context.

The physical data model is “derived” from the logical data model and may represent the same information (it can also be a subset of a logical data model) but simplified and enhanced for ease of use, efficiency and maintainability purposes.

DWH Migrations – PDM

Recommendation

- Data warehouse migrations migrate an existing data warehouse model from one technology to another. This of course means that technical considerations that decided the physical model are inherited.
- Considering that the DWH PDM (Core Integration Layer) primarily supports ETL tasks there's a tradeoff between efficiency and familiarity / experience. In most cases there is no huge benefit of remodeling existing PDM's for efficiency. However, if more work is necessary such as DWH consolidation it makes sense to re-assess the PDM based on the new technology.

Summary

Logical Data Modeling models data driven facts about how the business is operated without consideration of how the data will be used.

- LDM leads to stable, flexible, and unambiguous definitions since it only concerns itself with how the business is conducted.
- It establish the foundation for efficient communication around core business concepts staying true to the business context.

Physical Database modelling is modelling techniques to ensure:

- Correct
- Consistent
- Shareable
- Flexible

database like systems



Key takeaways

- Migrations often means accepting inheriting poor design choices for the new technology in favor to ensure familiarity and simplicity of migration.

Kimball Methodology

Kimballs 9 decision points when building a Data Warehouse

1. The business processes (the identity of the **fact** tables)
2. The **grain of each fact** table
3. The **dimensions of each fact** table
4. The facts including pre-calculated facts
5. The **dimension attributes** with descriptions and proper terminology
6. How to track **slowly changing dimensions**
7. The **aggregations**, type of dimensions, query modes and physical storage decisions
8. The **historical duration** of the database
9. The **urgency** with which data is extracted and loaded into the warehouse

Reference Data Management

What is Reference data?

Data characteristics

- Reference Data changes infrequently
- Reference Data is static in nature
- Reference Data serve as a constant reference point providing a common language for all systems within an organization eliminating ambiguity and discrepancy

By handling Reference Data systematically

- It can improve operational efficiency
- It may provide a common language for different systems
 - Particularly important for global organizations having localized definitions
- It plays a critical role in accurate and consistent reporting fostering comparability and meaningful interpretations
- It can play a big role in enforcing compliance

Pillars to be successful with RDM

Strategic plan

- Instituting a robust data governance model:
 - Framework for policies and procedures for how reference data is created, modified, stored, accessed and retired
- Shared understanding of reference data across all stakeholders

Harness the power of technology

- RDM oversees the whole lifecycle of reference data
- Initial collection, validation, storage, distribution, and retirement of reference data

Embracing Automation

- Data validation, consistency checks, sharing and monitoring use of reference data

ML and AI to extend Reference Data Management

In areas (example)

- Anomaly detection
- predictive analysis
- data validation
- reference data classifications

And generative AI can generate many of the reference data sets based on data models and usage, and highlight where gaps exist. To ensure accuracy of reference data and anticipate trends leading to more informed and strategic decisions regarding data quality and data vocabulary.

Issues if reference data is not dealt with

- Different systems using different lists (content or structure)
- Lists not fully aligned / used in different sub-organizations
- No clear definition and non unified encodings. A good example is that you may have a currency Swedish Krona, called SEK and that has a numeric encoding x

https://en.wikipedia.org/wiki/ISO_4217

https://en.wikipedia.org/wiki/ISO_10962

How MDM / RDM differs

Process

Master Data

1. Identify potential matches
2. Apply business rules and merge the records
3. Create Master record with validated attributes

Reference Data

1. Sources Data Fields
2. Provide lookups from Reference Table
3. Update sources to enterprise standard

Master Data Management Tools key features

Technology

- Out of the box support for multiple domains (customer, product and so on)
- Manage relationships between domains
- Categorization, grouping, and hierarchical management of entities
- Efficient and powerful Data deduplication algorithms to identify and remove duplicate records
- Configure elements required for matching according to business requirements
- Ability to define and execute business rules resulting in creation of a single golden record

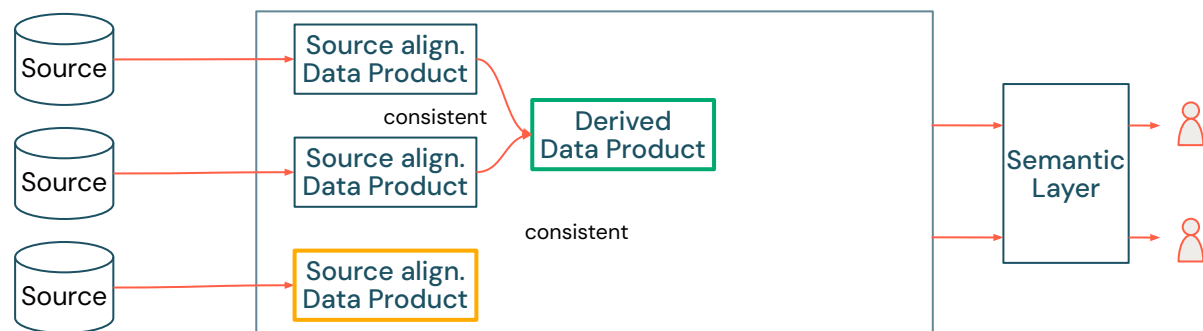
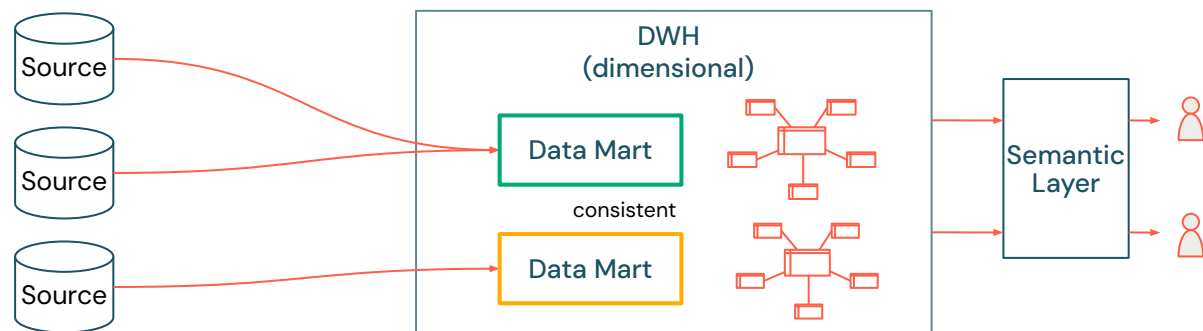
Logical Modeling

All the different data warehouse methodologies has parts that are about capturing the business context (logical data modeling) and parts that are related to efficiently organize data for answering questions on the data (physical data modeling).

Many times we forget the logical data modeling side since the physical data model is already available. However, the logical data model is what links the context of the business to the physical data model.

Attributes / benefits of each in a neutral way plus things to consider

Kimball vs. Data Products



Logical

Physical

